Signal Evaluation Metrics

The Relationship Between SNR and E_b/N_0

Signal-to-Noise Ratio (SNR) is the ratio of average signal power to average noise power in a given bandwidth.

$$\mathrm{SNR} = \frac{P_s}{P_n}$$

In digital communications, comparing systems based purely on SNR is inadequate because SNR depends on the specific bandwidth and data rate of the system. To establish a universal metric for system performance, we normalize power with respect to bit rate and bandwidth.

1. **Signal Power (P_s):** Can be expressed as the energy per bit (E_b) divided by the time duration of a single bit (T_b). Since bit rate (R_b) is the inverse of bit duration ($1/T_b$), we have:

$$P_s = \frac{E_b}{T_b} = E_b \cdot R_b$$

2. **Noise Power (P_n):** In an Additive White Gaussian Noise (AWGN) channel, noise power is the product of the noise power spectral density (N_0) and the system bandwidth (W).

$$P_n = N_0 \cdot W$$

Substituting these into the SNR equation yields:

$$\mathrm{SNR} = \frac{E_b \cdot R_b}{N_0 \cdot W}$$

Rearranging this gives the fundamental relationship:

$$\frac{E_b}{N_0} = \mathrm{SNR} \cdot \left(\frac{W}{R_b}\right)$$

E_b/N_0 is a dimensionless ratio (typically expressed in decibels, dB) that acts as the standard baseline for comparing the power efficiency of different digital modulation schemes, entirely independent of the physical bandwidth implementation.

The Origin of erf and erfc

In communication systems, thermal noise at the receiver is mathematically modeled as an Additive White Gaussian Noise (AWGN) process. This means the amplitude of the noise follows a Gaussian (Normal) probability distribution.

Bit Error Rate (BER) is the probability that the noise amplitude at the instant of sampling is large enough to push the received signal across a decision boundary, causing the receiver to misinterpret a `1` as a `0`, or vice versa.

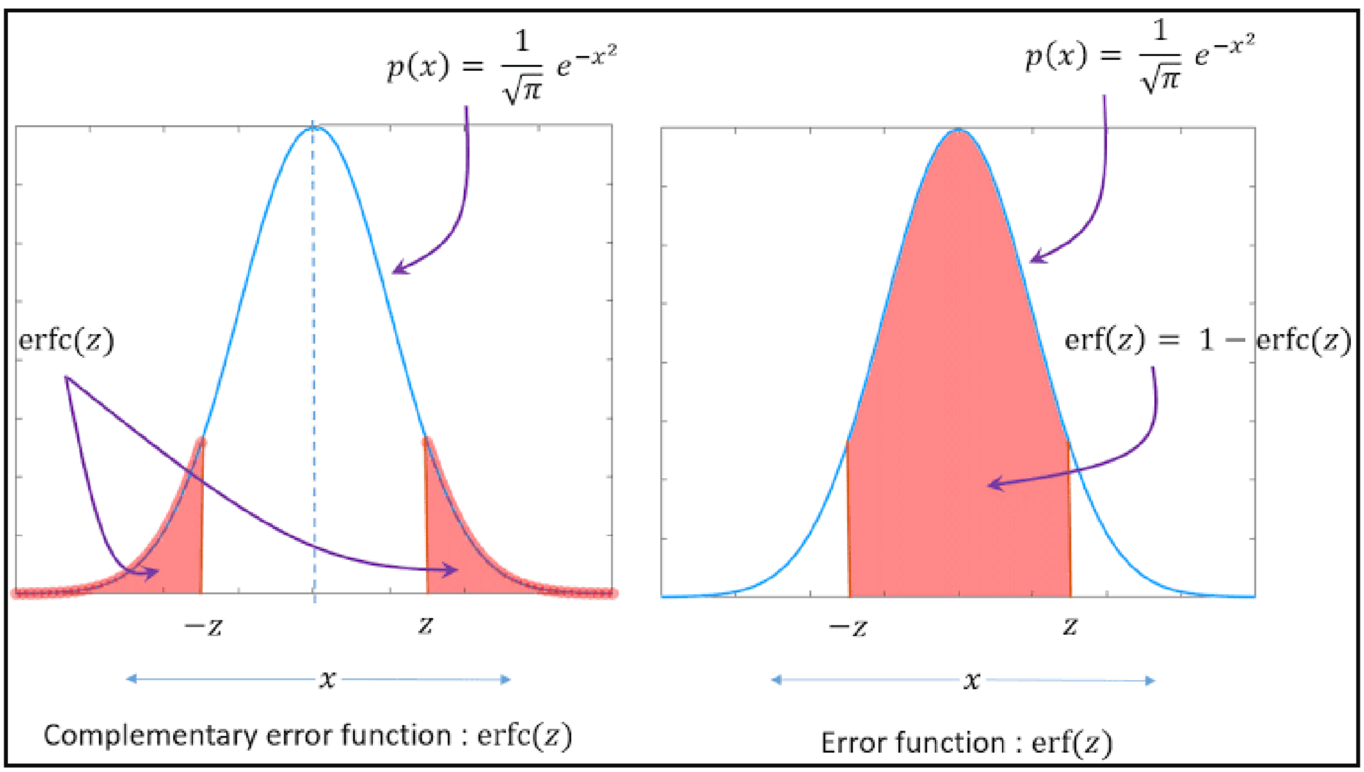
To calculate this probability, you must integrate the "tail" of the Gaussian Probability Density Function (PDF) starting from the decision boundary out to infinity.

The integral of the Gaussian PDF does not have a closed-form solution using elementary algebra. Therefore, the Error Function (erf) and the Complementary Error Function (erfc) are defined specifically to represent this integral.

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

$$\mathrm{erfc}(x) = 1 - \mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-t^2} dt$$

When you see erfc (or the closely related Q-function) in a BER formula, it is simply the exact mathematical representation of the area under the Gaussian curve that lies beyond the decision threshold.



The Formal Relationship: E_b/N_0 to BER

The relationship between E_b/N_0 and BER is theoretical and depends entirely on the **modulation scheme** you choose (e.g., BPSK, QAM, FSK) and the **channel model** (e.g., AWGN, Rayleigh fading).

As E_b/N_0 increases (meaning your signal is stronger compared to the noise), the BER decreases. In an ideal Additive White Gaussian Noise (AWGN) channel, the error probability P_b (which is the expected BER) for basic schemes like BPSK or QPSK is formally calculated using the Q-function:

$$P_b = Q\left(\sqrt{2\frac{E_b}{N_0}}\right) = \frac{1}{2}\text{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right)$$

(Note: The E_b/N_0 in this equation must be in linear scale, not dB).

When you plot this relationship on a graph—with E_b/N_0 (in dB) on the linear X-axis and BER on a logarithmic Y-axis—you get a **Waterfall-like Curve**.

Modulation	*DetectionMethod*	*BitErrorRate* (P_b)
BPSK	*Coherent*	$0.5erfc(\sqrt{\frac{E_b}{N_0}})$
QPSK	*Coherent*	$0.5erfc(\sqrt{\frac{E_b}{N_0}})$
M – PSK	*Coherent*	$\frac{1}{m}\, erfc\left(\sqrt{\frac{mE_b}{N_0}}sin(\frac{\pi}{M})\right)$
M – QAM(m = even)	*Coherent*	$\frac{2}{m}\left(1-\frac{1}{\sqrt{M}}\right)erfc(\sqrt{\frac{3mE_b}{2(M-1)N_0}})$
D – BPSK	*Non – coherent*	$0.5e^{-\frac{E_b}{N_0}}$
D – QPSK	*Non – coherent*	$Q_1(a,b) - 0.5I_0(ab)e^{-0.5(a^2+b^2)}$ where $a = \sqrt{\frac{2E_b}{N_0}\left(1-\frac{1}{\sqrt{2}}\right)}$ $b = \sqrt{\frac{2E_b}{N_0}\left(1+\frac{1}{\sqrt{2}}\right)}$ $Q_1(a,b)$ = Marcum Q -function $I_0(ab)$ = Modified Bessel-function

Why BER Formulas Differ for Phase-Modulation Schemes

BER is fundamentally governed by the Euclidean distance between adjacent symbols in a modulation scheme's constellation diagram (signal space).

For a constant average signal power P_s:

- In **BPSK** (Binary Phase Shift Keying), there are only two symbols separated by 180°. The distance between them is maximized, requiring a large noise vector to cause an error.
- In **QPSK** (Quadrature Phase Shift Keying), there are four symbols separated by 90°.
- In **8-PSK**, there are eight symbols separated by 45°.

As the modulation order increases to transmit more bits per symbol, the points in the constellation diagram are packed closer together. Because the decision boundaries are narrower, a much smaller noise vector can push the signal into the wrong decision region. Consequently, the limits of the Gaussian integral change, resulting in different analytical formulas and a higher BER for the same E_b/N_0 value.

Coherent vs. Non-Coherent Detection

This distinction refers strictly to whether the receiver utilizes the phase information of the incoming carrier wave.

Coherent Detection:

- The receiver possesses exact knowledge of the transmitter's carrier phase.
- It requires active phase-tracking circuitry, such as a Phase-Locked Loop (PLL), to generate a local oscillator signal that is perfectly phase-aligned with the incoming signal.
- **Advantage:** Because it projects the received signal perfectly onto the expected basis functions, it completely rejects noise that is orthogonal to the signal. This yields the theoretically optimal (lowest) BER.
- **Disadvantage:** High hardware complexity, cost, and acquisition time.

Non-Coherent Detection:

- The receiver operates without any knowledge of the absolute carrier phase.
- It relies on amplitude/envelope detection or compares the phase of the current symbol relative only to the *previous* symbol (Differential detection, e.g., DPSK).
- **Advantage:** Vastly simplified receiver architecture. No PLLs or phase synchronization required.
- **Disadvantage:** Because it cannot align exactly with the signal vector, it captures more noise power relative to the signal, resulting in a degradation of BER performance (typically requiring a higher E_b/N_0 to achieve the same BER as a coherent system).